

Sanjay Kumar Pilla

Software Engineer

MA, USA | (774) 525-7578 | sanjaypilla09@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Full Stack Developer with 2 years of experience in the IT industry, primarily on a finance project. Worked in Agile and Waterfall methodologies with strong expertise in .NET technologies including C#, multithreading, and collections. Applied design patterns and created design diagrams such as use case and sequence diagrams, while implementing architectures like MVC and Microservices. Skilled in .NET frameworks including ASP.NET Core Web Applications, ASP.NET Core Web API, ASP.NET MVC, and Razor Pages, with backend experience in building RESTful services using JSON and XML. Proficient in ORM frameworks such as ADO.NET, Entity Framework, and Entity Framework Core, with secure authentication using OAuth2, SSL, and JWT. Experienced in frontend development with Angular, TypeScript, HTML, CSS, JavaScript, Bootstrap, jQuery, and Ajax. Worked with SQL databases such as SQL Server and PostgreSQL, with exposure to NoSQL databases like MongoDB, and developed PL/SQL stored procedures, functions, and triggers. Performed testing using C# unit tests, Postman, and Swagger UI. Involved in CI/CD pipelines with TeamCity and Octopus, with cloud experience in AWS and GCP. Worked with version control tools including Git, and Bitbucket, using development tools such as Visual Studio, Visual Studio Code, and SQL Server Management Studio. Deployed applications on IIS, implemented logging with Log4Net, participated in code reviews, and used JIRA for project management, recognized as a quick learner and collaborative team player.

TECHNICALSKILLS

Languages:	C#, VB.NET, JavaScript, TypeScript, Python, SQL, HTML5, CSS3, Java, C, C++
.NET Technologies:	ASP.NET, ASP.NET Core, .NET Core, Razor Pages, Entity Framework, ADO.NET
Web Technologies:	Angular, React.js, Node.js, Express.js, Bootstrap, jQuery, Ajax, REST APIs, GraphQL, WebSocket's, Redux, Responsive Design, Cross-Browser Compatibility
Databases:	SQL Server, PostgreSQL, MySQL, Oracle, Snowflake, MongoDB
ORM Frameworks:	Entity Framework, Entity Framework Core, ADO.NET
Web Servers:	IIS, Nginx
Testing Frameworks:	C# Unit Testing, Postman, Swagger UI
Web Services:	RESTful APIs, API Design & Integration, OAuth2, JWT, Authentication & Authorization, Microservices
Version Control Tools:	Git, GitHub, Bitbucket
Methodologies:	SDLC, Agile (Scrum, Kanban), Waterfall, CI/CD Practices, Test-Driven Development (TDD), Behavior-Driven Development (BDD), DevOps, Code Reviews
Cloud Technologies:	Amazon Web Services (EC2, S3, RDS, Lambda, CloudWatch), Google Cloud Platform (Compute Engine, Cloud Storage, BigQuery, Cloud Functions), Serverless Architecture, Docker, Kubernetes, Logging & Monitoring (ELK Stack, Prometheus)
Other Skills:	Power BI, Tableau, Looker, QlikView, SSIS, SSRS, DBT, Azure Data Factory, ETL Pipelines, Data Warehousing, Data Integration, Data Pipeline Development, Data Transformation, Data Storytelling, System Design (MS Visio), Performance Optimization, Application Security Best Practices, Webpack, Jira, Agile Boards, Artificial Intelligence, Machine Learning, Deep Learning, Data Preprocessing, Feature Engineering, Model Training & Evaluation, NLP, Computer Vision, TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Data Visualization, Microsoft Visual Studio, Visual Studio Code

PROFESSIONAL EXPERIENCE

Software Engineer | Nagarro, India

January 2022 – November 2023

- Led end-to-end full-stack development of complex, enterprise-grade finance applications using .NET (C#, ASP.NET, .NET Core, Razor Pages), Angular, and MySQL, delivering highly performant, scalable, secure, and robust features that aligned with client requirements, enhanced user satisfaction, ensured overall system reliability, and adhered to coding standards and best practices.
- Designed and implemented robust, secure, and highly scalable RESTful APIs and backend services using .NET Core, enabling seamless integration with front-end applications in Angular. Performance optimizations and efficient database interactions reduced response times by up to 40%, significantly enhancing overall system efficiency, reliability, maintainability, and end-user experience across multiple platforms.
- Built interactive, responsive UI components with Angular, focusing on usability, accessibility, and cross-browser compatibility. These enhancements improved user engagement, client satisfaction, and overall experience across platforms and devices.
- Optimized complex SQL queries and redesigned database schemas, improving data retrieval speed and supporting complex business logic. Achieved a 25% increase in application performance and smoother transaction processing.
- Utilized CI/CD pipelines in TeamCity and Octopus to deploy applications, following automated workflows and best practices. Ensured smooth, reliable, and timely releases while improving deployment consistency and minimizing errors.
- Participated in Agile ceremonies such as daily standups, sprint planning, and retrospectives, actively managing tasks and user stories in JIRA to ensure timely feature delivery and effective collaboration within cross-functional teams.
- Participated in code reviews and implemented unit testing, ensuring high coding standards, security protocols, regulatory compliance, and maintainable, secure, and scalable codebases.
- Integrated Git version control via Bitbucket across distributed teams, enabling efficient collaboration, smooth code merges, and reliable rollback during deployments and bug fixes.
- Applied SOLID design principles and clean code practices to build modular, maintainable software systems, minimizing technical debt and ensuring scalability and adaptability to evolving business needs.
- Identified system bottlenecks and implemented process improvements, enhancing development efficiency, code quality, and consistent delivery of reliable software solutions.

PROJECTS

Reimbursement Portal

Technologies: Angular, .NET Core, SQL Server, Entity Framework Core, RESTful APIs

- Developed a web-based reimbursement portal using Angular and ASP.NET Core Web API, enabling submission of expense claims, approvals, and secure receipt management in SQL Server.
- Designed and implemented RESTful APIs with Entity Framework Core, including validation, error handling, and logging for reliable backend operations.
- Integrated role-based access control for different user types, ensuring secure access to features.
- Built front-end forms with Angular, including dynamic validations, responsive design, and interactive dashboards.
- Deployed the application on AWS and managed code with Git.

Book Reading Event

Technologies: ASP.NET Core, Razor Pages, Entity Framework, SQL Server

- Developed an application to manage global book-reading events, supporting public and private events, user registration, event invitations, and role-based access control with an admin user.
- Designed the frontend with Razor Pages and HTML/CSS/JavaScript/Bootstrap, implementing intuitive navigation, past/future event listing, event details, and “My Events” and “Events Invited To” views for logged-in users.
- Implemented backend using ASP.NET Core Web API and Entity Framework, including server-side validation, authentication (Authenticate, Authorize), authorization, and database logging for all commands.
- Created unit tests for core business methods, used interceptors to log database interactions, and ensured secure and maintainable event management functionality.

Temples Portal

Technologies: AngularJS, JSON Server

- Developed a web application for booking darshan, sevas, accommodations, and donations at Indian temples with secure user authentication.
- Built the frontend with AngularJS, creating responsive UI components, dynamic forms, and smooth navigation to enhance user experience.
- Integrated with JSON Server to simulate backend data storage for temples, bookings, and donations using JSON format.
- Practiced Angular concepts like two-way data binding, services, directives, and component-based architecture for skill improvement.

Sales Insights Dashboard

Technologies: Tableau, Snowflake

- Developed an interactive Sales Insights Dashboard to visualize brand, product, and regional sales performance using Tableau and datasets from Snowflake.
- Created key visualizations including bar charts, line charts, scatter plots, and maps, and implemented dynamic filtering and parameters to enable flexible data exploration.
- Provided actionable insights for trend prediction, customer segmentation, and regional sales strategy, supporting data-driven decision-making.

Movie Recommendation System Using Autoencoders

Technologies: Python, Pandas, NumPy, PyTorch, Matplotlib, MovieLens Dataset

- Implemented a deep autoencoder-based recommender system to predict unrated movies and generate personalized recommendations based on user rating history.
- Built encoder-decoder architecture trained with Masked Mean Squared Error (MMSE) loss, accurately reconstructing user-item rating matrices.
- Preprocessed datasets trained the model on cloud (Colab), visualized training and validation loss curves, and produced top-N movie recommendations tailored to each user.

EDUCATION

Master of Science in Computer Science - Clark University, Worcester, Massachusetts, USA

Bachelor of Technology – Lovely Professional University, Punjab, India